

“EUREKA! I HAVE FOUND IT.”

Attributed to Archimedes, Greek inventor and philosopher c.287–c.212 BCE

The Roman writer Vitruvius recorded that when Archimedes got into his bath he noticed that his body displaced a certain amount of water. This gave him the idea for the Archimedes Principle.

MAGNETIC IRON LODESTONES

were described in Chinese literature of the 3rd century BCE. By c.83 the Chinese text *Lun-heng* (*Discourses Weighed in the Balance*) had mentioned the electrostatic qualities of amber, which becomes charged when rubbed.

At around this time, Chinese diviners may also have discovered that iron, when rubbed against a lodestone, **becomes magnetized** and will point in a particular direction. The first primitive

compasses were iron ladles set on divining boards that pointed south.

In Greece, **Theophrastus of Lesbos** (c.370–287 BCE), a pupil of Aristotle and also his successor as head of the Lyceum school in Athens, **extended Aristotle's work**, particularly in botany. He wrote *Enquiry into Plants* and *On the Causes of Plants*, which **classified plants into trees, shrubs, and herbs**. He also began the study of plant reproduction and discussed

the best methods of cultivation for agriculture and companion planting to combat pests.

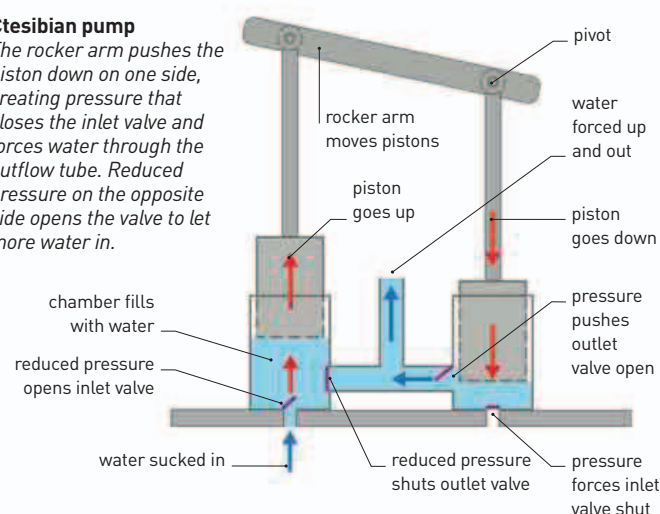
In astronomy, **Aristarchus of Samos** (c.310–230 BCE) rejected the prevailing view among early Greek astronomers that Earth was at the center of the Solar System. He believed that **Earth rotated in orbit around the Sun**; whether he thought the other planets also orbited the Sun is unclear. Aristarchus estimated the **comparative sizes** of the Sun and Earth at a ratio of about

20:1, and calculated the distance between Earth and the Sun to be 499 times the radius of Earth.

The science of **pneumatics** was founded by **Ctesibius of Alexandria** in the early 3rd century BCE. It is said that one of his first inventions was

Ctesibian pump

The rocker arm pushes the piston down on one side, creating pressure that closes the inlet valve and forces water through the outflow tube. Reduced pressure on the opposite side opens the valve to let more water in.



an adjustable-height mirror for his father's barber shop that used air compressed by counterweights to move up and down. He developed this idea to produce the **Ctesibian device**, a two-chamber force pump that used pistons attached to a rocker to create pressure. With the chambers of the device immersed in water, the rocker was moved up and down, alternately sucking water into one chamber and forcing it out of the other.

Another inventor and philosopher, **Archimedes** (287–212 BCE) was also one of the greatest mathematicians of Ancient Greece. In *On the Measurement of a Circle* he presented a method for **calculating the area and**

circumference of a circle. He also produced methods for calculating the **volumes of solids**, proving that the volume of a sphere inside a circumscribed cylinder is two-thirds that of the cylinder. Archimedes was the founder of hydrostatics (the science of fluids at rest). He showed that objects placed in water will displace a quantity of liquid equal to their buoyancy. He also developed a systematic **theory of statics**, showing how two weights balance each other at distances proportional to their relative magnitude. His aptitude for practical applications led him to develop the Archimedes screw (see 700–400 BCE) to pump out the bilges of a huge ship he built for the ruler of Syracuse. During



Chinese compass
A Han-era compass in the form of a magnetized ladle set on a bronze plate, featuring a diviner's representation of the cosmos.

c.300 BCE
Theophrastus of Lesbos starts to classify plants

c.250 BCE
Aristarchus of Samos develops heliocentric theory

c.260 BCE
Archimedes works on calculating the circumference and area of a circle

c.250 BCE
Ctesibius of Alexandria develops the Ctesibian pump